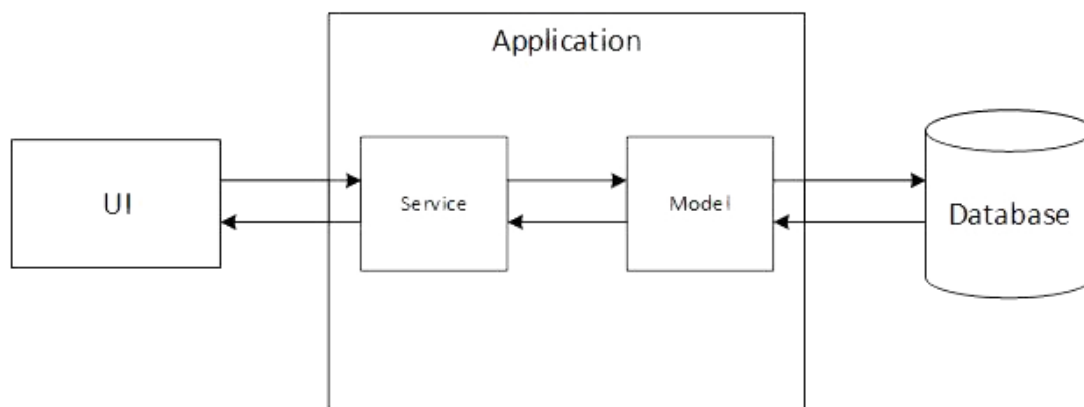


The Command Query Responsibility Segregation pattern

Command Query Responsibility Segregation (CQRS) is a pattern in which the model that is used to read information is separated from the model that is used to update information. In a more traditional architecture, a single object model is used for both reading and updating data:



Compromises become necessary in order to use a single object model as domain classes are required to serve all purposes. The same representation of an entity must support all of the **create, read, update, and delete (CRUD)** operations, making them larger than they need to be in all circumstances.

They contain all of the properties the object will need for various scenarios. If the class is more than just a **data transfer object (DTO)**, it may also contain methods for behavior. With this approach, classes are not ideal for all of the situations in which they need to be used as there is often a mismatch between what is required by the read and write representations of the data. This can also make managing security and authorization more complex as each class is used for both read and write operations.

In a collaborative domain, multiple operations may be taking place in parallel on the same set of data. There is a risk of data contention if records are locked, or update conflicts due to concurrent updates. Work-

loads between read and write tasks differ, which means that they also have different performance and scalability requirements.

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